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Transformers and BERT

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15 June 2026

- Introduction
- (Recap) Neural Networks for Encoders
- The Transformer
 - Attention
 - Multi-Head Attention
 - Positional encoding
 - Add and Normalize
- Tokenization
- Transformer-Encoder Models
 - BERT
 - Applied fine-tuning and evaluation
 - RoBERTa
 - ModernBERT



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Section 1

Introduction



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Roadmap

By the end, you should be able to:

- Distinguish encoder-only, decoder-only, and encoder-decoder transformer models

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- Distinguish encoder-only, decoder-only, and encoder-decoder transformer models
- Explain self-attention in words and connect the intuition to the formula



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- Distinguish encoder-only, decoder-only, and encoder-decoder transformer models
- Explain self-attention in words and connect the intuition to the formula
- Understand BERT pretraining, fine-tuning, and different types of classification



By the end, you should be able to:

- Distinguish encoder-only, decoder-only, and encoder-decoder transformer models
- Explain self-attention in words and connect the intuition to the formula
- Understand BERT pretraining, fine-tuning, and different types of classification
- Know common research-design pitfalls when using transformer encoders in social science

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The Development of NLP

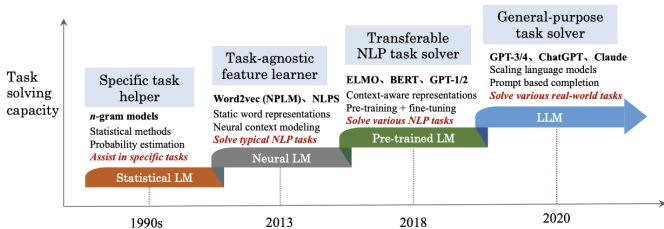


Figure: The development timeline (Zhao et al., 2023)



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- Learning feature representations



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- Learning **feature representations**
- Good for "sensor" data



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- Learning **feature representations**
- Good for "sensor" data
- Needs a lot of data to learn **complex representations** (image, text, audio)



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Learning Representations

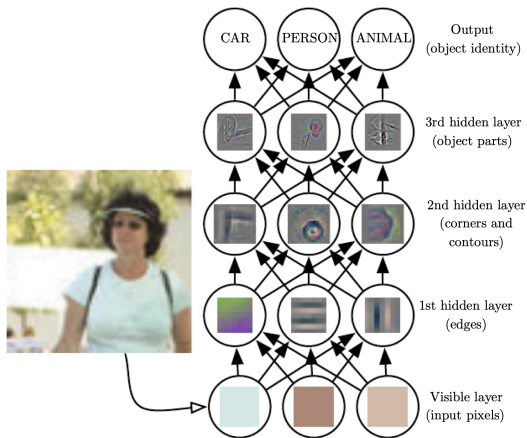


Figure: Learning representations can be crucial (Goodfellow et al, 2017, Fig. 1.2)



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- Different networks for different purposes

- **Feed-Forward Neural Network:** Basic building block



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- Different networks for different purposes

- **Feed-Forward Neural Network:** Basic building block
- **Convolutional Neural Networks:** Computer Vision



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- Different networks for different purposes

- **Feed-Forward Neural Network:** Basic building block
- **Convolutional Neural Networks:** Computer Vision
- **Transformers:** Textual data



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Encoder-only

BERT, RoBERTa,
ModernBERT

- Classification
- Retrieval
- Extraction
- Measurement
- Bidirectional context

Decoder-only

GPT, Claude

- Generation
- Chat
- Open-ended reasoning
- Causal context: previous tokens only



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Section 2

(Recap) Neural Networks for Encoders



Feed-forward neural network

For binary classification, a one-hidden-layer feed-forward network can be written as:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1),$$

$$\eta = \mathbf{w}_2^\top \mathbf{h} + b_2,$$

$$\hat{y} = \frac{1}{1 + \exp(-\eta)}.$$

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$$\hat{y} = \frac{1}{1 + \exp(-\eta)}.$$

Example:

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{W}_1 = \begin{bmatrix} 0.5 & -1 \\ 1 & 0.5 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{z}_1 = \begin{bmatrix} -1.5 \\ 2 \end{bmatrix}, \quad \mathbf{h} = \text{ReLU}(\mathbf{z}_1) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\eta = [1 \quad -0.5] \mathbf{h} + 0.3 = -0.7$$

$$\hat{y} = \frac{1}{1 + \exp(0.7)} \approx 0.33.$$



Common non-linearities

- **ReLU**: keeps positive activations and clips negative values

$$\text{ReLU}(x) = \max(0, x)$$

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Common non-linearities

- **ReLU**: keeps positive activations and clips negative values

$$\text{ReLU}(x) = \max(0, x)$$

- **GeLU**: a smooth gate that weights x by how likely it is under a standard normal

$$\text{GeLU}(x) = x\Phi(x) \approx \frac{1}{2}x \left(1 + \tanh \left[\sqrt{\frac{2}{\pi}}(x + 0.044715x^3) \right] \right)$$

- Unlike ReLU, GeLU does not hard-zero all negative inputs
- Smooth gating can give more stable gradients and often works well in Transformer feed-forward layers

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- Unlike ReLU, GeLU does not hard-zero all negative inputs
- Smooth gating can give more stable gradients and often works well in Transformer feed-forward layers
- **GeGLU**: splits a layer into content and gate parts

$$\text{GeGLU}(\mathbf{a}, \mathbf{b}) = \mathbf{a} \odot \text{GeLU}(\mathbf{b})$$

where $\odot$ indicates element-wise multiplication.



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- If layers were only linear/affine:

$$\mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} + (\mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2)$$



Why non-linearities matter

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- So several stacked linear layers collapse into one linear layer



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- So several stacked linear layers collapse into one linear layer
- Non-linearities prevent that collapse and let depth build curved decision boundaries and interactions



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- A layer is a learned transformation of a (hidden) representation



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- A layer is a learned transformation of a (hidden) representation
- Non-linearities such as ReLU, GeLU, and GeGLU prevent stacked layers from collapsing into one linear map



What we need from neural networks

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- An **output head** maps a representation to a prediction: class, token label, score, or embedding
- Fine-tuning updates the encoder and/or head to reduce a loss function



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The Transformer



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Attention is all you need.



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 - Enables more GPU **parallelism**



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 - Enables deeper networks for text



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Attention is all you need.
- Behind the recent progress in NLP: BERT, Claude, GPT, etc.
- Benefits for textual data:
 - Enables more GPU *parallelism*
 - Better handling of *long-range dependencies*
 - Enables *transfer learning* for text data
 - Enables *deeper* networks for text
- I will rely on images from
Alammar (2018) *The Illustrated Transformer*
(recommended)



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A Sequence-to-Sequence Model



Figure: The basic block (Alammar, 2018)



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Stacked Encoder-Decoder Structure

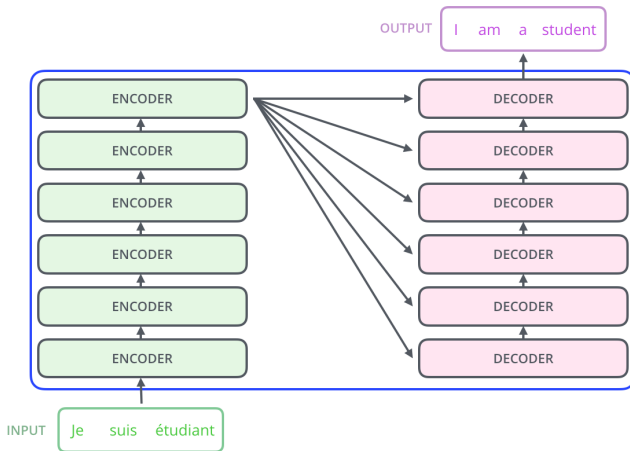


Figure: The Transformer layers (Alammar, 2018)



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Transformer

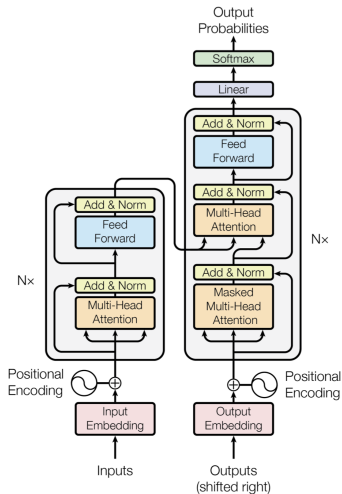


Figure: The Transformer Architecture (Vaswani et al., 2017)



The encoder vs. the decoder

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- Encoder:
 - Input: tokens/subword tokens
 - Output: **contextualized** embeddings



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- Encoder:
 - Input: tokens/subword tokens
 - Output: **contextualized** embeddings
- Decoder:
 - Input: **previous tokens** (and contextualized embeddings from encoder)
 - Output: next word prediction



Masking determines what the model can see

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- **BERT-style encoders:** each token can attend to tokens on both sides



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- **BERT-style encoders:** each token can attend to tokens on both sides
- **GPT-style decoders:** each token can only attend to previous tokens



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- **BERT-style encoders:** each token can attend to tokens on both sides
- **GPT-style decoders:** each token can only attend to previous tokens
- This is why encoders are useful for classification, extraction, and measurement



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- **BERT-style encoders:** each token can attend to tokens on both sides
- **GPT-style decoders:** each token can only attend to previous tokens
- This is why encoders are useful for classification, extraction, and measurement
- It is also why GPT-style decoder models are natural next-token generators

Example: in Equality Act 2010, the token Act can use Equality and 2010 when deciding whether it is part of a legal citation.



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The Transformer Layer (Encoder layer)

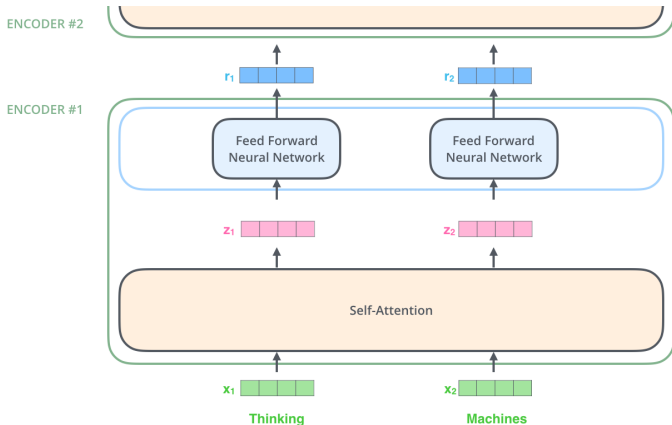


Figure: The Encoder Layer (Alammar, 2018b)



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Self-attention intuition

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- Start with a token: Act



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- Start with a token: Act
- Its query asks: which other tokens help interpret me?



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- Start with a token: Act
 - Its query asks: which other tokens help interpret me?
 - Other tokens return keys: Equality, 2010, and nearby words may match strongly



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- Start with a token: Act
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- Start with a token: Act
- Its query asks: which other tokens help interpret me?
- Other tokens return keys: Equality, 2010, and nearby words may match strongly
- Their values pass information into the representation of Act
- After several layers, every token representation is contextualized by the context around the words



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Scaled Dot-Product Attention

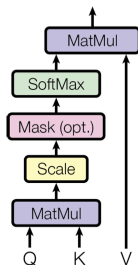


Figure: Scaled Dot-Product Attention (Vaswani et al., 2017)

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} \right) \mathbf{V}$$

- Query: what this token is looking for



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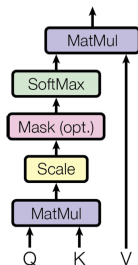


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- Query: what this token is looking for
- Key: what other tokens offer



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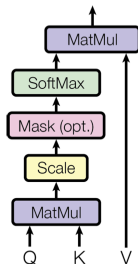


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- Query: what this token is looking for
- Key: what other tokens offer
- Value: the information passed into the new representation



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Computing Q, V and K

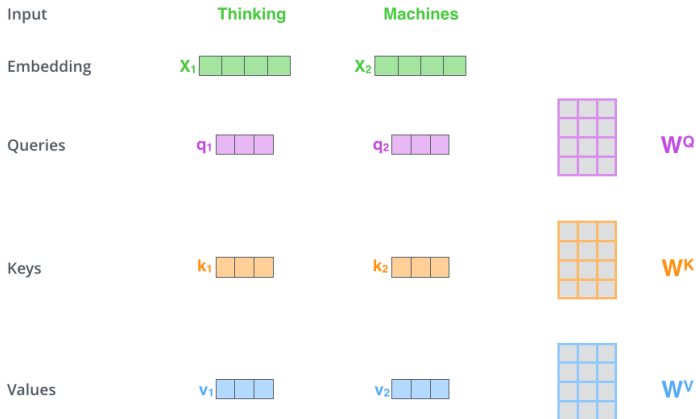


Figure: Attention heads (Alammar, 2018b)



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Computing Self-Attention

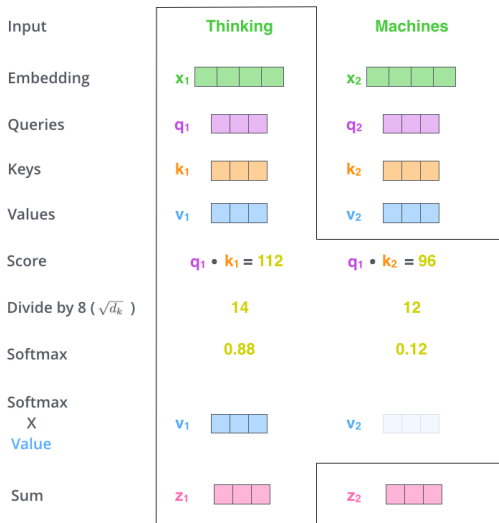


Figure: Attention (Alammar, 2018b)



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Multi-Head Attention

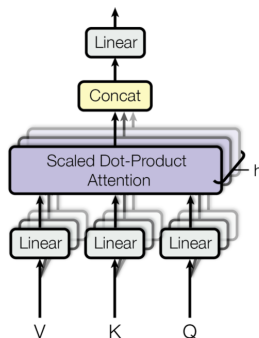


Figure: Multi-Head Attention (Vaswani et al., 2017)



Attention Heads

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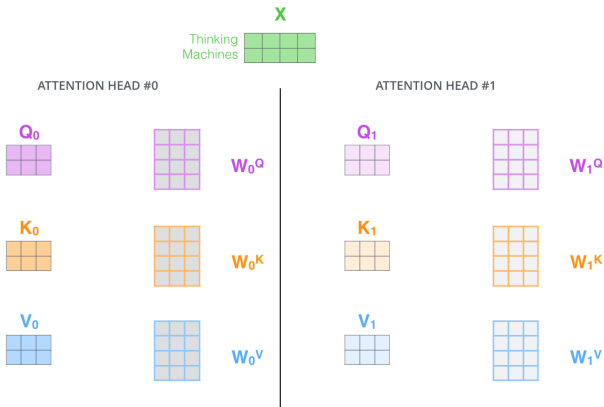


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Multi-head attention

- 1) This is our input sentence*
- 2) We embed each word*
- 3) Split into 8 heads. We multiply X or R with weight matrices
- 4) Calculate attention using the resulting $Q/K/V$ matrices
- 5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer

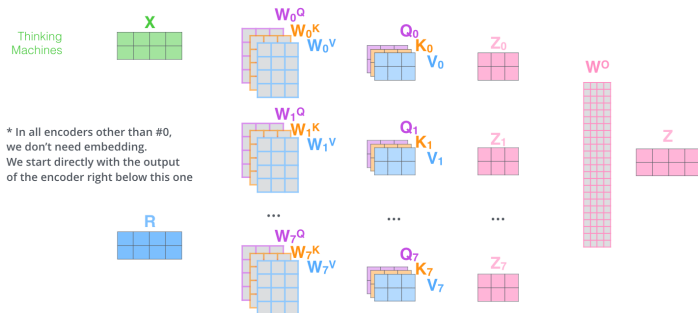


Figure: Positional encoding vectors (Alammar, 2018b)



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Multi-Head Attention example

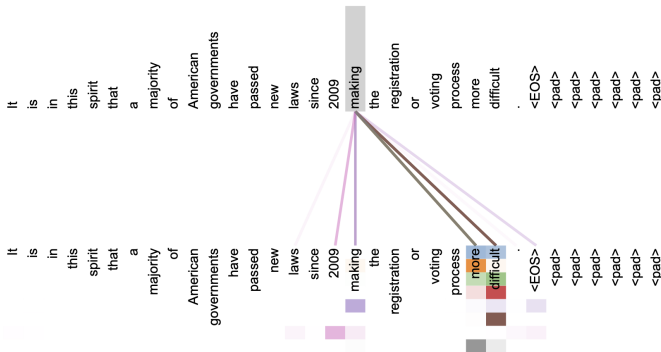


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb ‘making’, completing the phrase ‘making...more difficult’. Attentions here shown only for the word ‘making’. Different colors represent different heads. Best viewed in color.

Figure: Attention (Vaswani et al., 2017)



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Positional Encoding

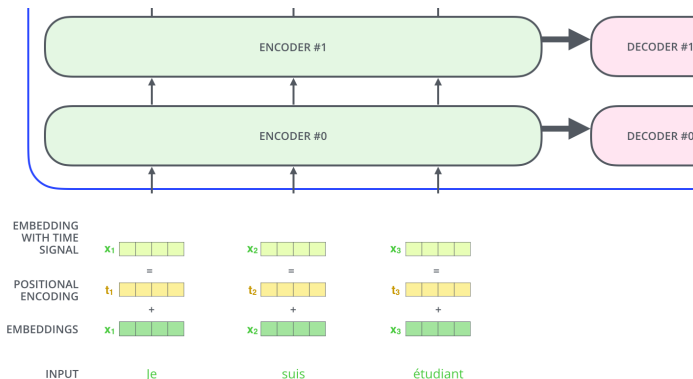


Figure: Attention heads (Alammar, 2018b)



(Absolute) Positional Encoding

"The boy hit the ball" vs "The ball hit the boy"

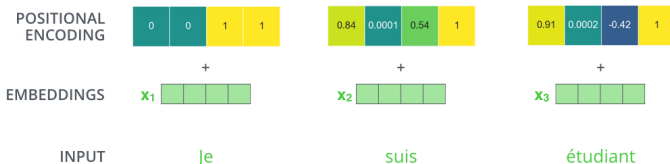


Figure: Adding positional encodings to embeddings (Alammar, 2018b)

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"The boy hit the ball" vs "The ball hit the boy"

- **Absolute:** add a position vector to each token vector

$$\mathbf{e}_i = \mathbf{x}_i + \mathbf{p}_i$$

Token at position 2 gets $\mathbf{p}_2$, regardless of what it attends to.

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"The boy hit the ball" vs "The ball hit the boy"

- **Absolute:** add a position vector to each token vector

$$\mathbf{e}_i = \mathbf{x}_i + \mathbf{p}_i$$

Token at position 2 gets $\mathbf{p}_2$, regardless of what it attends to.

- **Relative:** add information about the distance between two tokens to the attention score

$$s_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d}} + b_{i-j}$$

The pair (2, 5) and the pair (7, 10) both have distance +3.

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RoPE intuition

Rotary Position Embedding (RoPE) puts position into the query and key vectors before the dot product.

$$\tilde{\mathbf{q}}_i = \mathbf{R}_i \mathbf{q}_i, \quad \tilde{\mathbf{k}}_j = \mathbf{R}_j \mathbf{k}_j$$

For intuition, look at one two-dimensional rotation:

$$\mathbf{R}_i = \mathbf{r}(i\theta), \quad \mathbf{R}_j = \mathbf{r}(j\theta), \quad \mathbf{r}(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

Here θ is the rotation rate per token position. Position i rotates by $i\theta$, position j by $j\theta$; comparing them sees the relative angle $(j - i)\theta$. Full RoPE repeats this with several θ values.

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- Absolute encoding says: “this token is at position i ”
- Relative encoding says: “token j is $j - i$ steps away”
- RoPE rotates $\mathbf{q}_i$ and $\mathbf{k}_j$ so their dot product depends on the relative offset $j - i$

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- RoPE rotates $\mathbf{q}_i$ and $\mathbf{k}_j$ so their dot product depends on the relative offset $j - i$
- Why use it? It gives relative-position behavior without adding a separate learned position table, and it is efficient for longer contexts

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Add and Normalize

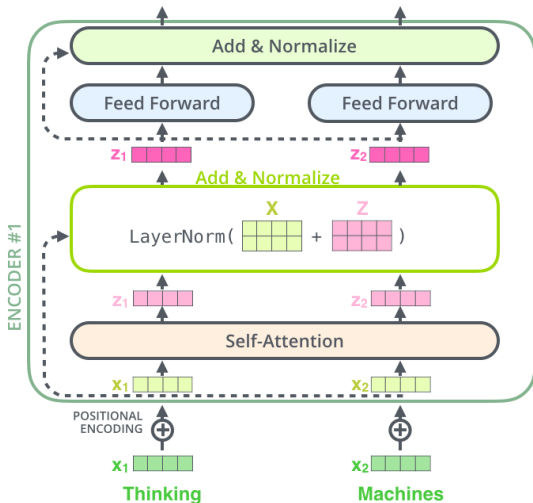


Figure: Add and Normalize (Alammar, 2018b)



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Transformer

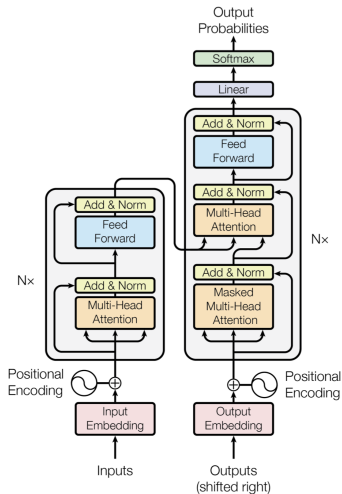


Figure: The Transformer Architecture (Vaswani et al., 2017)



Tokenization

- Models do not read words directly; they read token IDs from the model's vocabulary

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- Models do not read words directly; they read token IDs from the model's vocabulary
- Subword tokenization balances three problems:
 1. Very large vocabulary size
 2. Out-of-vocabulary (OOV) tokens
 3. Different meanings of very similar words



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prefer pairs with high $\frac{P(i,j)}{P(i)P(j)}$



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prefer pairs with high $\frac{P(i,j)}{P(i)P(j)}$

- Practical issue: subword labels and long chunks must be mapped back to text



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Section 5

Transformer-Encoder Models



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Transformer-Encoder Models

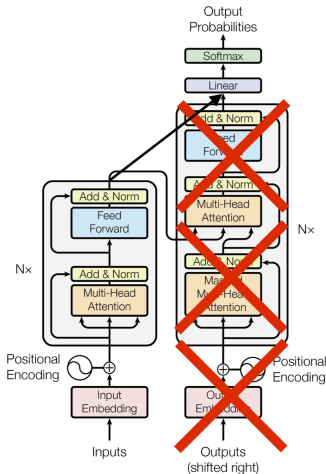


Figure: The Transformer Architecture (Vaswani et al., 2017)



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Transformer-Encoder Models

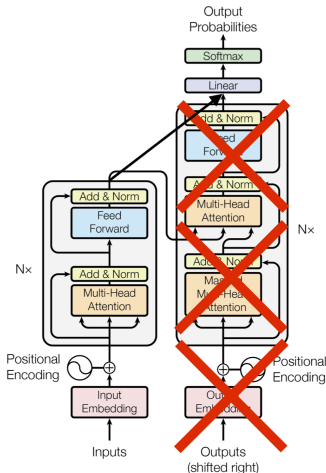


Figure: The Transformer Architecture (Vaswani et al., 2017)

- Common models are BERT, RoBERTa, and ModernBERT



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- Bidirectional Encoder Representations from Transformers (BERT)
- Introduced in Devlin et al. (2018)



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 - Named-Entity Recognition
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- Many flavors, such as RoBERTa, ALBERT, etc.



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- Bidirectional Encoder Representations from Transformers (BERT)
- Introduced in Devlin et al. (2018)
- Introduced **state-of-the-art** performance in many text prediction tasks, such as
 - Named-Entity Recognition
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- Many flavors, such as RoBERTa, ALBERT, etc.
- **Pre-trained** on a large corpus



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- And again, I rely on Alammr (2018) **The Illustrated BERT**



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Why BERT still matters in 2026

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- They return stable labels, scores, spans, and embeddings that are easy to evaluate
- They are efficient when we need to process thousands or millions of documents



Where BERT-style encoders are used today

BERT here means the broader encoder family: BERT, RoBERTa, ModernBERT, and embedding-focused encoders.

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- Same backbone, different head or loss: similarity loss for embeddings, cross-entropy for classification, sequence labeling loss for spans



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- Document classification: [CLS] or pooled vector → document label



Same encoder, different output heads

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- Document classification: [CLS] or pooled vector → document label
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- **Document classification**: [CLS] or pooled vector → document label
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- **Document classification**: [CLS] or pooled vector → document label
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- **Retrieval**: document and query vectors → similarity score
- **Measurement**: predictions become variables (prediction-powered inference)

Example: detecting legal citations in UK court judgments as token classification.



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BIO2 is a span-labeling scheme: B- begins an entity, I- continues it, and O means outside any entity.

Token	The	Equality	Act	2010	applies	.
BIO2	O	B-STATUTE	I-STATUTE	I-STATUTE	O	O



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The span is reconstructed from consecutive B-/I- labels:

Equality Act 2010



Encoders often sit around a decoder LLM in a larger pipeline.

- **Retrieval**: embed documents and find passages to give the LLM as context

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How encoders complement LLMs

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- **Retrieval**: embed documents and find passages to give the LLM as context
- **Routing**: classify a case and send it to the right prompt, model, or tool
- **Filtering**: remove irrelevant documents before the expensive LLM call
- **Quality control**: check whether an LLM output matches expected labels, spans, or policy rules

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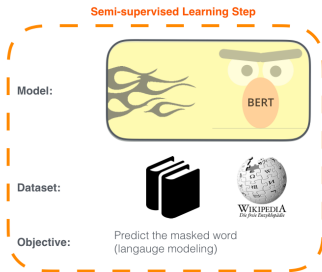


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BERT and transfer learning

1 - **Semi-supervised** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



2 - **Supervised** training on a specific task with a labeled dataset.

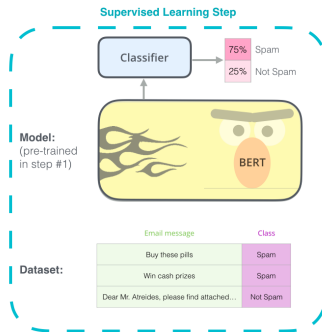


Figure: Using BERT for Transfer Learning (Alammar, 2018b)



Domain adaptation: do not stop pretraining

Domain adaptation means continuing self-supervised pretraining on text closer to the target problem before supervised fine-tuning.

general encoder → domain/task text with MLM →
classifier or extractor

- **DAPT**: continue pretraining on unlabeled in-domain text, such as legal judgments, clinical notes, or financial reports

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- **TAPT**: continue pretraining on the unlabeled texts from the target task itself
- Why it helps: the model sees domain vocabulary, style, genres, and cues before learning from scarce labels
- Modern recipe: DAPT/TAPT when the domain differs, then full fine-tuning for the supervised task

Sun et al. (2019) emphasized further pretraining before fine-tuning; Gururangan et al. (2020) formalized DAPT/TAPT and showed gains across domains and tasks.



text $\rightarrow$ tokenizer $\rightarrow$ encoder $\rightarrow$ contextual vectors
 $\rightarrow$ task head $\rightarrow$ supervised loss

- **Document classification**: one vector, usually [CLS], predicts a document label

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- The model is only as good as the annotation scheme, split strategy, and evaluation design



The BERT model

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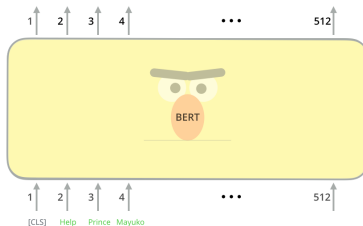


Figure: The BERT model (Alammar, 2018b)



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BERT Architecture

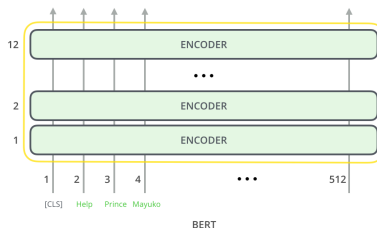


Figure: Opening up BERT (Alammar, 2018b)



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Training Task 1: Masked Language Model

Use the output of the masked word's position to predict the masked word

Possible classes:
All English words

0.1%	Aardvark
...	...
10%	Improvisation
...	...
0%	Zzyzyva

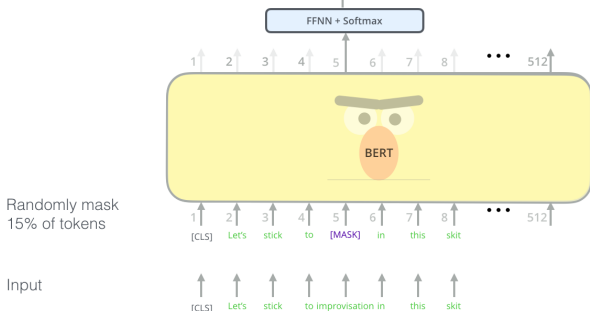


Figure: Masked Language Modeling (Alammar, 2018c)



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Original BERT task 2: Next Sentence Prediction

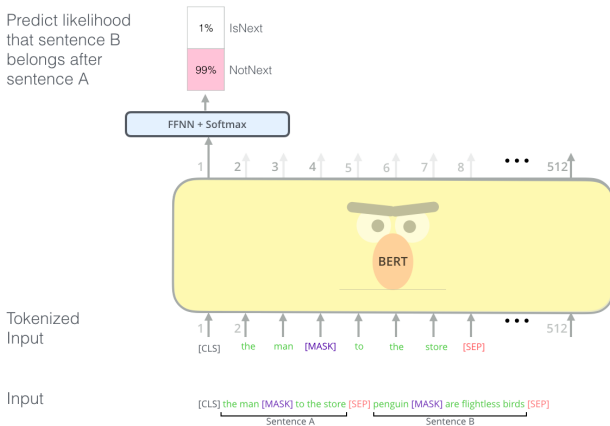


Figure: Next Sentence Prediction (Alammar, 2018c)

Historical note: this was part of original BERT, but later models such as RoBERTa removed the NSP objective.



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Using BERT for Classification

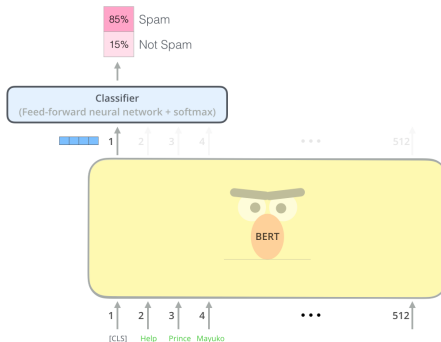


Figure: Using BERT for classification (Alammar, 2018c)



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BERT and Contextualized embeddings

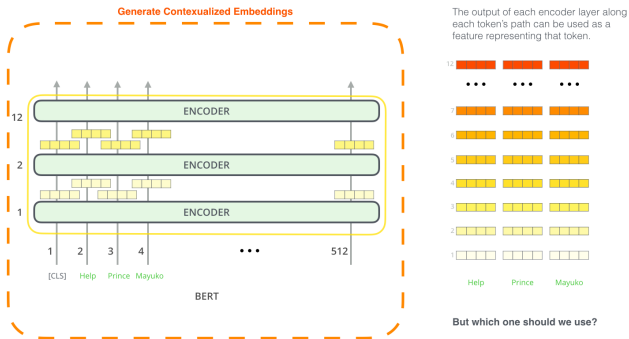


Figure: Contextualized Embeddings (Alammar, 2018c)



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Using Contextualized Embeddings

What is the best contextualized embedding for “Help” in that context?
For named-entity recognition task CoNLL-2003 NER

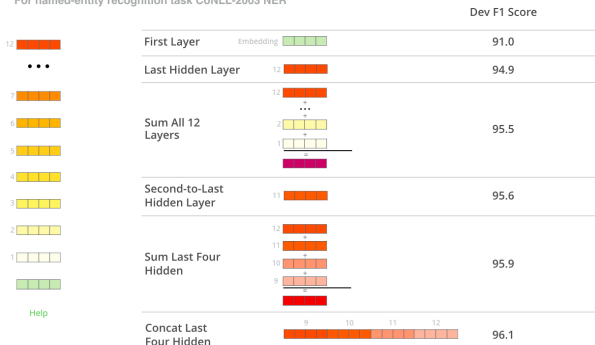


Figure: Using Contextualized Embeddings (Alammar, 2018c)



Example: legal citation detection

Sargeant, Östling, and Magnusson (2025) detect legal citations in UK court judgments.

- **Task:** identify references to legislation and case law in text

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- **Why it is useful:** citation networks, precedent mapping, retrieval, and legal measurement



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Method	F1	Precision	Recall
Regex, strongest baseline	35.42	72.98	23.38
GPT-4.1, dynamic few-shot	76.57	63.36	96.74
BERT-cased, fine-tuned	92.50	92.44	92.56
LEGAL-BERT, fine-tuned	92.72	89.74	95.92
ModernBERT, fine-tuned	93.30	92.73	93.88

Results from the all-label test set in Sargeant, Östling, and Magnusson (2025).



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Encoder fine-tuning beats shortcuts

Method	F1	Precision	Recall
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Results from the all-label test set in Sargeant, Östling, and Magnusson (2025).

Lesson: for high-volume extraction, supervised encoders can be more accurate, cheaper, and easier to evaluate than regex or prompted decoder models.



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- Split by document/case, not by sentence chunks that leak context across splits



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- Report token-level and span/entity-level performance
- Inspect precision and recall separately: false positives and false negatives have different research costs
- Audit errors by label, period, court, genre, language, and subgroup where relevant



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Research pitfalls

- Domain shift: a model trained on one corpus may fail on another

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- Class imbalance: rare labels can disappear behind strong aggregate F1



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- Max sequence length and chunking: long documents can lose context at boundaries
- Calibration and bias: prediction scores are not automatically valid measurements



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- Better pretraining recipe: dynamic masking, no next sentence prediction, more data, longer training, larger batches



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- Same encoder-only idea as BERT
- Better pretraining recipe: dynamic masking, no next sentence prediction, more data, longer training, larger batches
- Takeaway: RoBERTa showed that the pretraining setup mattered a lot; it was not a fundamentally new architecture



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ModernBERT: a modern encoder

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- Warner et al. (2025): Smarter, Better, Faster, Longer



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- Trained on roughly 2 trillion tokens with a native 8,192-token sequence length



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- Same broad use case as BERT: encoder-only representations for discriminative tasks
- Trained on roughly 2 trillion tokens with a native 8,192-token sequence length
- Strong on classification and retrieval, including long-context retrieval
- Useful for social-science workflows that require scalable labeling and embedding



What changed since BERT?

- Longer context: 8,192 tokens instead of the classic 512-token limit

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- Faster attention: alternating local/global attention plus Flash Attention support
- Less wasted compute: unpadding for variable-length batches
- Modern architecture/data choices: GeGLU layers, hardware-aware design, larger and more diverse pretraining data

Caveat: ModernBERT is still an encoder; it does not replace decoder LLMs for open-ended generation.